

Short Communication

Ruthenium/NHC-catalyzed tandem benzylic oxidation/oxidative esterification of benzylic alcohols with phenols

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ABSTRACT

An efficient methodology to access benzoate derivatives via tandem benzylic oxidation/oxidative esterification of benzylic alcohols with phenols catalyzed by ruthenium/NHC was developed. This operationally simple one-pot process uses O₂ as the clean oxidant, producing esters in good to excellent yields.

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1. Introduction

Ester groups are among the highly important and abundant functional groups in organic chemistry. Thus, it is worthwhile to provide an environmentally benign method to access esters [1,2]. The existent traditional methodology involves the activation of the carboxylic acid as an acyl halide, anhydride, or activated ester followed by nucleophilic substitution [3–8]. From the atom-economic point of view, the direct oxidative esterification of aldehydes remains as an attractive possibility to readily access esters [9–20]. Recently, the transition-metal-catalyzed tandem oxidation of alcohol to aldehyde followed by an oxidative esterification has received much attention [21–27]. For example, Scheidt demonstrated a tandem oxidation of alcohols to esters using N-heterocyclic carbenes as catalysts, which required super-stoichiometric amounts of MnO₂ as the oxidant [28]. Recently, Lei and Beller developed palladium-catalyzed aerobic oxidative esterification of benzylic alcohols, respectively [29,30]. However, less attention has been paid to the synthesis of aryl benzoate derivatives, which were building blocks of numerous active compounds and cross coupling partners [31,32]. Very recently, Chen developed a palladium/NHC-catalyzed tandem benzylic oxidation/oxidative esterification of benzylic alcohols with phenols to access aryl benzoate derivatives [33]. Nevertheless, a facile and versatile procedure on such transformation still remains a highly desired goal for organic chemists. Herein, we report a ruthenium/NHC-catalyzed tandem oxidation/esterification of benzylic alcohols with phenols, using O₂ as the clean terminal oxidant.

Table 1

Selected results of screening the optimal conditions.

Entry	[Ru]	Ligand	Base	Solvent	Yield (%) ^a
1	[Ru(p-cymene)Cl ₂] ₂	–	Cs ₂ CO ₃	Xylene	Trace
2	[Ru(p-cymene)Cl ₂] ₂	L1	Cs ₂ CO ₃	Xylene	8
3	[Ru(p-cymene)Cl ₂] ₂	L2	Cs ₂ CO ₃	Xylene	20
4	[Ru(p-cymene)Cl ₂] ₂	L3	Cs ₂ CO ₃	Xylene	61
5	[Ru(p-cymene)Cl ₂] ₂	L4	Cs ₂ CO ₃	Xylene	80
6	[Ru(p-cymene)Cl ₂] ₂	L5	Cs ₂ CO ₃	Xylene	90 (Trace) ^b
7	[Ru(p-cymene)Cl ₂] ₂	L5	Cs ₂ CO ₃	Xylene	(76) ^c (<5) ^d
8	[Ru(p-cymene)Cl ₂] ₂	L5	Na ₂ CO ₃	Xylene	53
9	[Ru(p-cymene)Cl ₂] ₂	L5	K ₂ CO ₃	Xylene	40
10	[Ru(p-cymene)Cl ₂] ₂	L5	<i>t</i> -BuOK	Xylene	11
11	[Ru(p-cymene)Cl ₂] ₂	L5	–	Xylene	63
12	[Ru(p-cymene)Cl ₂] ₂	L5	Cs ₂ CO ₃	1,4-Dioxane	61
13	[Ru(p-cymene)Cl ₂] ₂	L5	Cs ₂ CO ₃	DCE	Trace
14	[Ru(p-cymene)Cl ₂] ₂	L5	Cs ₂ CO ₃	Toluene	30
15	[Ru(p-cymene)Cl ₂] ₂	L5	Cs ₂ CO ₃	DMF	<5
16	[Ru(p-cymene)Cl ₂] ₂	L5	Cs ₂ CO ₃	Xylene	85 ^e
17	[Ru(p-cymene)Cl ₂] ₂	L5	Cs ₂ CO ₃	Xylene	46 ^f
18	[Ru(p-cymene)Cl ₂] ₂	L5	Cs ₂ CO ₃	Xylene	62 ^g
19	Ru ₃ (CO) ₁₂	L5	Cs ₂ CO ₃	Xylene	70
20	RuCl ₂ (PPh ₃) ₃	L5	Cs ₂ CO ₃	Xylene	85
21	RuCl·3H ₂ O	L5	Cs ₂ CO ₃	Xylene	12

^aAll reactions were run with phenylmethanol **1a** (0.3 mmol), 4-nitrophenol **2a** (0.2 mmol), [Ru] (5 mol %), ligand (10 mol %) and base (10 mol %), in dry solvent (2 mL) under O₂ at 130 °C for 24 h. Isolated yields: ^bin the absence of Ru, ^cunder air, ^dunder N₂, ^eCs₂CO₃ (20 mol %), ^fCs₂CO₃ (50 mol %), and ^gat 110 °C.

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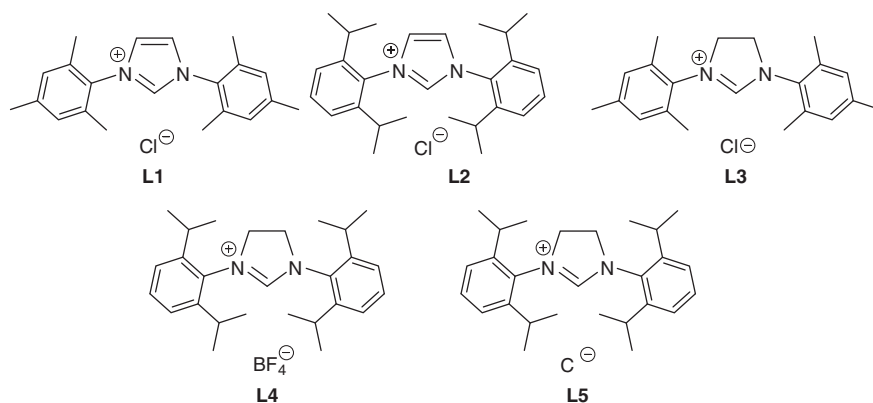


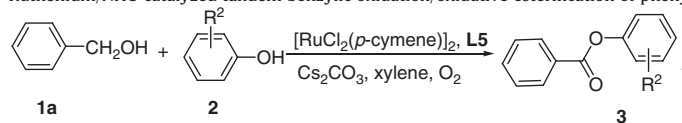
Fig. 1. Selected imidazolium salts screened.

2. Results and discussion

From the economical and environmental point of view, the use of molecular oxygen as the terminal oxidant has attracted considerable attention. With this in mind, we began our investigation using

phenylmethanol with 4-nitrophenol as the model reaction employing $[\text{RuCl}_2(p\text{-cymene})]_2$ the catalyst under oxygen (Table 1). Only a trace of the product was formed in the absence of ligand as determined by GC–MS (Table 1, entry 1). Various precursors of N-heterocyclic carbene (NHC) (Fig. 1) were investigated for this

Table 2
Ruthenium/NHC-catalyzed tandem benzylic oxidation/oxidative esterification of phenylmethanol with phenols.



Entry	Phenol 2	Product 3	Yield (%) ^a
1	2a	3aa	88
2	2b	3ab	83
3	2c	3ac	95
4	2d	3ad	82
5	2e	3ae	80
6	2f	3af	85
7	2g	3ag	79
8	2h	3ah	90
9	2i	3ai	81
10	2j	3aj	84
11	2k	3ak	90

^aReaction conditions: **1a** (0.3 mmol), **2** (0.2 mmol), $[\text{RuCl}_2(p\text{-cymene})]_2$ (5 mol %), L5 (10 mol %) and Cs_2CO_3 (10 mol %) in dry xylene (2 mL) under O_2 at 130 °C for 24 h. Isolated yield.

reaction and an excellent yield was obtained with L5 (Table 1, entry 6). No product was detected by GC–MS when the reaction was carried out under nitrogen and 76% yield was provided under air condition (Table 1, entry 7). These results suggested that the oxygen played a significant role in the reaction. Meanwhile, the base used also had an important effect in the reaction and Cs_2CO_3 proved to be the best. In addition, the amount of Cs_2CO_3 had dramatic effect

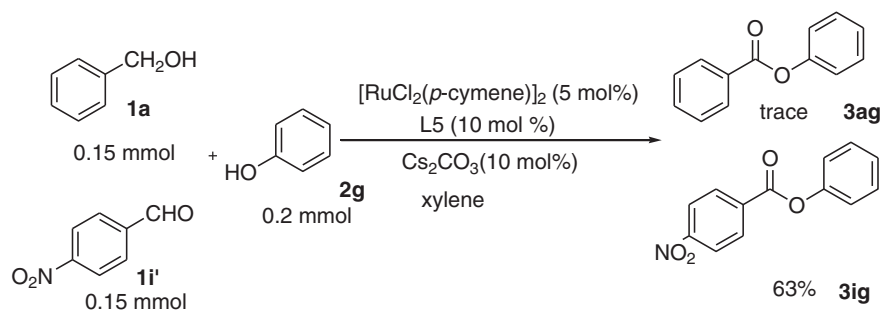
on the reaction (Table 1, entries 6, 16 and 17). For example, the yield was sharply decreased to 46% when using 50 mol % of Cs_2CO_3 . Moreover, reaction in dioxane, DCE, DMF and toluene resulted in low efficiencies or no reaction (Table 1, entries 12–15). Further studies showed that the $[\text{RuCl}_2(p\text{-cymene})]_2$ was the best ruthenium source (Table 1, entries 3 and 19–21). The yield was sharply decreased to 62% when the procedure ran at 110 °C. In

Table 3

Ruthenium/NHC-catalyzed tandem benzylic oxidation/oxidative esterification of phenylmethanol with phenols.

Entry	Substrate 1	Product 3	Yield (%) ^a
1			
1			86
2			60
3			84
4			79
5			77
6			86
7			88
8			60
9			83
10			86
11			76
12			90
13			65
14			52
15			50

^aReaction conditions: **1** (0.3 mmol), **2** (0.2 mmol), $[\text{RuCl}_2(p\text{-cymene})]_2$ (5 mol %), L5 (10 mol %) and Cs_2CO_3 (10 mol %) in dry xylene (2 mL) under O_2 at 130 °C for 24 h. Isolated yield.



Scheme 1. Competitive reaction of phenol with phenylmethanol and 4-nitrobenzaldehyde.

addition, only a trace of the product was detected by GC–MS in the absence of $[\text{RuCl}_2(p\text{-cymene})]_2$.

With the optimized conditions in hand, the scope of the reaction with respect to phenols was investigated (Table 2). The reaction proceeded smoothly and tolerated various functional groups such as methoxy, nitro, cyano, chloro, acetamido, fluoro and bromo groups and provided the products in good to excellent yields. The substituent on different positions on the aromatic ring of phenols affected the reaction yield slightly. Importantly, the reaction tolerates halogens groups (Table 2, entries 4–6) and the surviving halogen atom ($\text{X} = \text{Cl}, \text{Br}$) could be valuable for further manipulation.

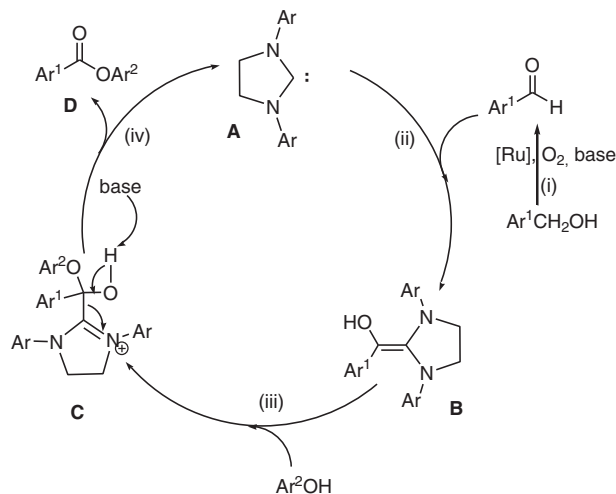
Next, the scope of the esterification of various primary alcohols was examined (Table 3). The strong electron-withdrawing groups such as $-\text{CF}_3$, $-\text{COOMe}$ and $-\text{NO}_2$ could also be tolerated under the standard conditions. In case of allyl alcohol and 1-butanol as allylic alcohol substrates, the desired ester was isolated in 52% and 50% yield, respectively (Table 3, entries 14–15).

To understand the mechanism more clearly, more experiments were carried out. Conversion of benzylic alcohol to the aromatic aldehyde was observed under the standard procedure by GC–MS. Moreover, under the standard procedure without L5, naphthalene-1-methanol produced 1-naphthaldehyde in 93% yield (Scheme 1). Next, when 1-naphthaldehyde and benzaldehyde were subjected to the standard procedure, the desired products were isolated in 75% and 85% yield, respectively (Scheme 2). These results could disclose the possibility of a tandem oxidation of benzylic alcohol to aromatic aldehyde followed by an esterification pathway.

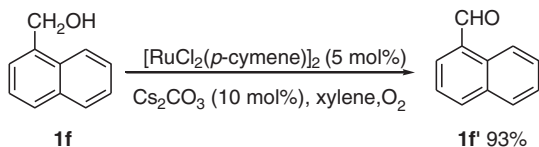
Furthermore, a competitive experiment using phenol with alcohol and aldehyde was conducted (Scheme 3). A mixture with 1:1 ratio of **1a** and 4-nitrobenzaldehyde **1i'** with **2g** was carried out under the same conditions as for entry 6, Table 1. A trace of **3ag** was detected, while **3ig** was isolated in 63% yield after 6 h. This result suggests that this reaction involves the oxidation of alcohols to

aldehydes as the key rate-determining step in the course of the reaction.

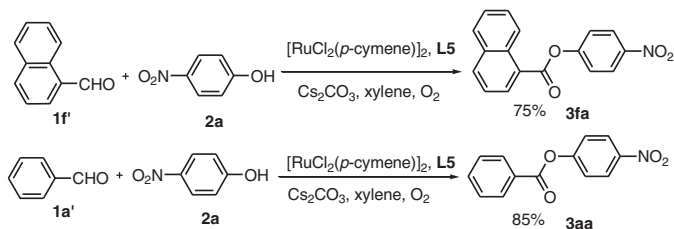
A plausible mechanism for this tandem oxidation/esterification is outlined in Scheme 4. First, aromatic aldehyde was produced from benzylic alcohols via ruthenium-catalyzed aerobic oxidation reaction [34]. In step (ii), the deprotonation of the heteroazolium salt generates the N-heterocyclic carbene **A** that adds to the carbonyl of the acyl group, after which, the Breslow's intermediate undergoes an oxidation step to afford **B** [16,28,35]. In step (iii), the reaction of **B** with phenol produces intermediate **C**. Ultimately, in step (iv), intermediate **C** produces ester **D** and regenerates NHC catalyst to complete the catalytic cycle in the presence of base [16,28,35].



Scheme 4. Plausible mechanism.



Scheme 2. Oxidation of naphthalene-1-methanol to 1-naphthaldehyde.



Scheme 3. Oxidative esterification of aryl aldehydes with phenols.

In summary, we have developed an efficient methodology to access benzoate derivatives via tandem benzylic oxidation/oxidative esterification of benzylic alcohols with phenols catalyzed by ruthenium/NHC. Notably, the amount of base in the procedure is only 10 mol %. This operationally simple one-pot process uses O_2 as the clean oxidant, producing esters in good to excellent yields.

3. Experimental

Chemicals were either purchased or purified by standard techniques without special instructions. ^1H NMR and ^{13}C NMR spectra were measured on a 500 MHz spectrometer (^1H 500 MHz, ^{13}C 125 MHz), using CDCl_3 as the solvent with tetramethylsilane (TMS) as the internal standard at room temperature. Chemical shifts

(δ) are given in ppm relative to TMS; the coupling constants J are given in Hz.

3.1. General procedure of ruthenium/NHC-catalyzed tandem benzylic oxidation/oxidative esterification of benzylic alcohols with phenols

Under oxygen, a reaction tube was charged with phenols (0.2 mmol), benzylic alcohols (0.3 mmol), $[\text{RuCl}_2(p\text{-cymene})]_2$ (6.1 mg, 5 mol %), L5 (8.5 mg, 10 mol %), Cs_2CO_3 (6.5 mg, 10 mol %) in dry xylene (2 mL). After the mixture was stirred for 24 h at 130 °C, the solvent was evaporated under reduced pressure and the residue was purified by flash column chromatography on silica gel to give the product.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at doi:10.1016/j.catcom.2011.12.041.

References

- [1] J. Otera, Esterification: Methods, Reactions, and Applications, Wiley-VCH, Weinheim, 2003.
- [2] A. Parenty, X. Moreau, J.-M. Campagne, Chem. Rev. 106 (2006) 911–939.
- [3] K. Ishihara, Tetrahedron 65 (2009) 1085–1109.
- [4] J. Otera, Acc. Chem. Res. 37 (2004) 288–296.
- [5] N. Nowrouzi, A.M. Mehranpour, J.A. Rad, Tetrahedron 66 (2010) 9596–9601.
- [6] G.A. Grasa, R. Singh, S.P. Nolan, Synthesis (2004) 971–985.
- [7] T. Shinada, M. Hamada, K. Miyoshi, M. Higashino, T. Umezawa, Y. Ohfuné, Synlett (2010) 2141–2145.
- [8] M. Hatano, Y. Furuya, T. Shimmura, K. Moriyama, S. Kamiya, T. Maki, K. Ishihara, Org. Lett. 13 (2011) 426–429.
- [9] S.D. Sarkar, S. Grimme, A. Studer, J. Am. Chem. Soc. 132 (2010) 1190–1191.
- [10] K. Zeitler, Angew. Chem., Int. Ed. 44 (2005) 7506–7510.
- [11] S.-I. Kiyooka, Y. Wada, M. Ueno, T. Yokoyama, R. Yokoyama, Tetrahedron 63 (2007) 12695–12701.
- [12] S. De Sarkar, A. Biswas, C.H. Song, A. Studer, Synthesis (2011) 1974–1983.
- [13] B. Maji, S. Vedachalan, X. Ge, S. Cai, X.-W. Liu, J. Org. Chem. 76 (2011) 3016–3023.
- [14] H. Jiang, B. Gschwend, L. Albrecht, K.A. Jørgensen, Org. Lett. 12 (2010) 5052–5055.
- [15] M. Zhang, S. Zhang, G. Zhang, F. Chen, J. Cheng, Tetrahedron Lett. 52 (2011) 2480–2483.
- [16] R.S. Reddy, J.N. Rosa, L.F. Veirois, S. Caddick, P.M.P. Gois, Org. Biomol. Chem. 9 (2011) 3126–3129.
- [17] J.N. Rosa, R.S. Reddy, N.R. Candeias, P.M.S.D. Cal, P.M.P. Gois, Org. Lett. 12 (2010) 2686–2689.
- [18] Z. Ye, G. Lv, W. Wang, M. Zhang, J. Cheng, Angew. Chem., Int. Ed. 49 (2010) 3671–3674.
- [19] F. Luo, S. Pan, C. Pan, P. Qian, J. Cheng, Adv. Synth. Catal. 353 (2011) 320–324.
- [20] G. Lv, G. Huang, G. Zhang, C. Pan, F. Chen, J. Cheng, Tetrahedron 67 (2011) 4879–4886.
- [21] T. Zweifel, J.-V. Naubron, T. Büttner, T. Ott, H. Grützmaier, Angew. Chem., Int. Ed. 47 (2008) 3245–3248.
- [22] H. Miyamura, T. Yasukawa, S. Kobayashi, Green Chem. 12 (2010) 776–778.
- [23] M.G. Coleman, A.N. Brown, B.A. Bolton, H. Guana, Adv. Synth. Catal. 352 (2010) 967–970.
- [24] N. Yamamoto, Y. Obora, Y. Ishii, J. Org. Chem. 76 (2011) 2937–2941.
- [25] M. Ito, A. Osaku, A. Shiibashi, T. Ikariya, Org. Lett. 9 (2007) 1821–1824.
- [26] Y. Obora, Y. Ishii, Synlett (2011) 30–51.
- [27] C. Gunanathan, L.J.W. Shimon, D. Milstein, J. Am. Chem. Soc. 131 (2009) 3146–3147.
- [28] B.E. Maki, A. Chan, E.M. Phillips, K.A. Scheidt, Org. Lett. 7 (2007) 371–374.
- [29] C. Liu, J. Wang, L. Meng, Y. Deng, Y. Li, A. Lei, Angew. Chem., Int. Ed. 50 (2011) 5144–5148.
- [30] S. Gowrisankar, H. Neumann, M. Beller, Angew. Chem., Int. Ed. 50 (2011) 5139–5143.
- [31] A. Moretto, A. Nicoli, M. Lotti, Toxicol. Appl. Pharm. 219 (2007) 196–201.
- [32] M.D. Barratt, D.A. Basketter, D.W. Roberts, Toxicol. in Vitro 8 (1994) 823–826.
- [33] F. Luo, C. Pan, J. Cheng, F. Chen, Tetrahedron 67 (2011) 5878–5882.
- [34] M. Lee, S. Chang, Tetrahedron Lett. 41 (2000) 7507–7510.
- [35] P. Arde, B.T. Ramanjaneyulu, V. Reddy, A. Saxena, R. Vijaya Anand, Org. Biomol. Chem. 10 (2012) 848.